

Selective induction of apoptosis of human oral cancer cell lines by avocado extracts via a ROS-mediated mechanism.

Avocados have a high content of phytochemicals with potential chemopreventive activity. Previously we reported that phytochemicals extracted from avocado meat into a chloroform partition (D003) selectively induced apoptosis in cancer but not normal, human oral epithelial cell lines. In the present study, we observed that treatment of human oral cancer cell lines containing high levels of reactive oxygen (ROS) with D003 increased ROS levels twofold to threefold and induced apoptosis. In contrast, ROS levels increased only 1.3-fold, and apoptosis was not induced in the normal cell lines containing much lower levels of basal ROS. When cellular ROS levels in the malignant cell lines were reduced by N-acetyl-L-cysteine (NAC), cells were resistant to D003 induced apoptosis. NAC also delayed the induction of apoptosis in dominant negative FADD-expressing malignant cell lines. D003 increased ROS levels via mitochondrial complex I in the electron transport chain to induce apoptosis. Normal human oral epithelial cell lines transformed with HPV16 E6 or E7 expressed higher basal levels of ROS and became sensitive to D003. These data suggest that perturbing the ROS levels in human oral cancer cell lines may be a key factor in selective apoptosis and molecular targeting for chemoprevention by phytochemicals.